Greatest common factor

1 Find the greatest common factor between each set of terms:

a 3 and $15x$ **b** $8a$ and $9a$ **c** $33x$ and $6x^2$ **d** $12b$ and $15b^2x$

2 Complete the following statements:

a $4r + 12 = \square\,(r + 3)$ **b** $5v - 30 = \square\,(v - 6)$

c $18c - 24 = \square\,(3c - 4)$ **d** $2t^2 + 2t = 2t\,(\square + \square)$

3 Factor the following expressions:

a $6v + 30$ **b** $45t - 40$

c $-2s - 10$ **d** $-12s + 10$

e $y^2 + 4y$ **f** $k^2 - 5k$

g $2u^2 - 8u$ **h** $4t + 2t^2$

i $42x - x^2$ **j** $7x + 35$

k $36 - 6x$ **l** $-2x + 10$

m $-100 - 10x$ **n** $-18y^2 + 6$

o $9\,(3x + 4) + 4\,(3x + 4)$ **p** $8t\,(t - 3) + 9\,(t - 3)$

Grouping with 4 terms

4 Factor the following expressions:

a $5\,(a + b) + v\,(a + b)$ **b** $x\,(y - z) - w\,(y - z)$

c $8x\,(2y + 3w) - z\,(2y + 3w)$ **d** $8z\left(5x^2 + 4y\right) - \left(5x^2 + 4y\right)$

e $8y\,(y - 4) + 10\,(4 - y)$ **f** $(4c - d)\,(c + 9d) - 5\,(d - 4c)$

5 Determine whether or not the following expressions are in factored form:

a $-5\,(2y + z) - b\,(2y + z)$ **b** $(a + 7)\,(a + 2)$

c $(x - 5)\,(x + y)$ **d** $3x\,(a + 2b) + 2\,(a + 2b)$

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6 Factor the following expressions:

a $x^2 + 5x + 8x + 40$

b $x^2 - 5x + 10x - 50$

7 Factor the following expressions:

a $xy - 9x + 2y - 18$

b $56 - 8y - 7x + xy$

c $12xy + wx + 12yz + wz$

d $2x + xz - 40y - 20yz$

e $50 + 5y + 10x + xy$

f $12xy + wx + 12yz + wz$

8 By rewriting $4x^2 + 17x + 4$ as an expression having four terms and factoring in pairs, factor the expression completely.

Trinomials

9 Complete the expressions that will make the following equations true.

a $(m + 8)(\square) = m^2 + 18m + 80$

b $y^2 - 8y - 65 = (y - 13)(\square)$

10 Factor the following expressions:

a $x^2 + 6x + 5$

b $x^2 + 14x + 40$

c $x^2 - 5x + 4$

d $x^2 - 13x + 30$

e $x^2 - 15x + 56$

f $x^2 + x - 132$

g $x^2 + 9x - 90$

h $x^2 - 2x - 8$

i $x^2 - 3x - 70$

j $6 + 5x + x^2$

k $44 - 15x + x^2$

l $-12 - 13x - x^2$

m $-9 + 10x - x^2$

11 Factor the following expressions:

a $4x^2 + 56x + 192$

b $-5x^2 + 10x + 40$

c $3x^2 - 21x - 54$

d $2x^3 + 16x^2 + 30x$

12 Complete the following equations to show the correct factors of the given expressions.

a $3x^2 + 10x - 8 = (3x - 2)(x + \square)$

b $12x^2 + 25x + 7 = (4x + 7)(3x + \square)$

c $15x^2 - 13x - 6 = (5x - \square)(3x + \square)$

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13 Factor the following expressions:

a $7x^2 - 75x + 50$

b $6x^2 + x - 7$

c $9x^2 - 19x + 10$

d $7x^2 - 34x - 48$

e $6x^2 + 13x + 6$

f $-12x^2 - 7x + 12$

g $90 - 89x + 8x^2$

h $8 - 14p - 49p^2$

14 Factor the following expressions:

a $b^2 + 14b + 49$

b $y^2 - 10y + 25$

c $x^2 + 10x + 25$

d $x^2 + 14x + 49$

e $x^2 - 12x + 36$

f $x^2 - 16x + 64$

g $81x^2 - 72x + 16$

h $81x^2 + 36x + 4$

i $49x^2 - 28x + 4$

j $81 + 18x + x^2$

k $36 - 12x + x^2$

l $25 + 60r + 36r^2$

m $7d^2 + 112d + 448$

n $4x^2 + 40x + 100$

o $3x^2 - 24xy + 48y^2$

p $-6r^2 - 96r - 384.$

q $-3x^2 + 12x - 12$

r $b^2 - 7b + \frac{49}{4}$

s $x^2 + 3x + \frac{9}{4}$

t $x^2 - 3x + \frac{9}{4}$

Difference of squares

15 Write an expression that makes each of the following statements true.

a $p^2 - 64 = (p - 8)(\square)$

b $m^2 - 36 = (m - 6)\,(\square)$

16 Find the other factor of $x^2 - 49$ if one of its factors is $x + 7$.

17 Factor $x^4 - 100$ as a product of two binomials.

18 Consider the expression $81x^2 - 225$
Write the factored form of an equivalent expression, where a and b are integers.

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19 Factor the following expressions:

a $x^2 - 25$	**b** $n^2 - 36$	**c** $v^2 - 49$	**d** $x^2 - \frac{4}{49}$
e $81 - x^2$	**f** $4 - u^2$	**g** $4 - 49y^2$	**h** $64x^2 - 121$
i $121m^2 - 64$	**j** $81x^2 - 100y^2$	**k** $121x^2 - 49y^2$	**l** $16x^2 - 25$
m $6x^2 - 294$	**n** $3t^2 - 12$	**o** $5x^2 - 320$	**p** $4n^2 - 16$

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